CUSTOMER 360 AI

A PROJECT REPORT

Submitted by

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IN
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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **“Customer 360 AI: An Intelligent, Explainable Customer Analytics, Churn Prediction, and Retention Platform"** is the bonafide work carried out by **Abhishek Kumar (25MCI10327)** students of Master of Computer Applications (MCA), in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications from **Chandigarh University** during the academic year 2026–2027.

The project embodies original work carried out under appropriate academic supervision and has reached the required engineering standard for postgraduate submission.

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INTERNAL EXAMINER	EXTERNAL EXAMINER

DECLARATION OF ORIGINALITY

I, **Abhishek Kumar** student of Master of Computer Applications (MCA) at **Chandigarh University**, hereby declare that the project report titled **"Customer 360 AI: An Intelligent, Explainable Customer Analytics, Churn Prediction, and Retention Platform"** submitted to the Department of Computer Applications is a record of original work done by me under the supervision of **Rishabh Tomar**.

I confirm that:

1. This work has not been submitted in part or full for the award of any other degree, diploma, fellowship, or associate-ship at any other university or institution.
2. All data sources, theoretical formulations, literature citations, and third-party software libraries have been explicitly acknowledged according to standard academic conventions.
3. The screenshots and code presented represent genuine engineering implementations developed within the project repository.

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ACKNOWLEDGEMENTS

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ABSTRACT

In today's subscription- and e-commerce-driven economy, customer retention is the primary determinant of business longevity. Acquiring a new customer is widely documented to cost **5 to 7 times more** than retaining an existing buyer, and a modest 5% boost in customer retention can increase overall corporate profitability by 25% to 95%. However, contemporary Customer Relationship Management (CRM) and Business Intelligence (BI) platforms suffer from a pronounced **"Insight-to-Action Gap"**: they deliver retrospective historical metrics (what happened) while contemporary Machine Learning systems provide opaque churn scores (e.g., $P(\text{churn}) = 0.81$) without explaining underlying behavioral drivers or prescribing actionable retention interventions.

This project introduces **Customer360 AI**, an enterprise-grade, multi-tenant analytics and machine learning platform engineered around an explicit decision-support framework: **Facts → Predictions → Explanations → Recommendations → Simulations**.

The system integrates seven foundational engineering modules:

1. **Automated Feature Engineering & Data Quality Engine**: Translates multi-table relational transaction ledgers into standardized 10-dimensional Customer 360 vectors including Recency, Frequency, Monetary (RFM), Average Order Value (AOV), Purchase Velocity, and Customer Lifetime Value (CLV), evaluated with automated Data Quality (DQ) anomaly scoring.
2. **Dual-Tier Segmentation**: Couples classical rule-based RFM quantile mapping with unsupervised K-Means clustering, featuring automated optimal-k Silhouette selection and Principal Component Analysis (PCA) dimensionality reduction.
3. **Supervised Churn Modeling with Imbalance Compensation**: Implements stratified cross-validation across Logistic Regression, Random Forest, LightGBM, and XGBoost, addressing empirical class skew using cost-sensitive gradient scaling (`scale_pos_weight`) and selecting models based on PR-AUC and ROC-AUC.
4. **Game-Theoretic Explainability (XAI)**: Embeds TreeSHAP and LinearSHAP to compute both macro-level feature importance distributions and micro-level, local waterfall contribution charts for every single customer.
5. **Prescriptive Value-at-Risk Engine**: Maps customers into a 2×2 CLV $\times$ Churn Risk Matrix (Value-at-Risk = $P(\text{Churn}) \times \text{CLV}$), driving an 8-rule deterministic Next-Best-Action recommendation pipeline.
6. **AI Retention Advisor with AST-Grounded NL-to-SQL**: Enables non-technical managers to query the relational customer database in plain English, verified by an Abstract Syntax

Tree (AST) validator enforcing `SELECT`-only execution over a read-only database credential.

7. **Continuous Distribution Drift Monitoring**: Implements Population Stability Index (PSI) with sample-adaptive quantile binning alongside non-parametric two-sample Kolmogorov-Smirnov (KS) tests to detect feature and prediction covariate shift in production.

The architecture is implemented as a decoupled, high-performance web system comprising a **Python 3.12 / FastAPI** backend, **SQLAlchemy 2** ORM, **PostgreSQL 16**, **Redis 7** broker with **Celery** async workers, and a **React 18 / TypeScript / Vite / Tailwind CSS / Recharts** frontend. The final champion model achieves **89.7% Accuracy**, **91.8% ROC-AUC**, and **86.5% PR-AUC**, with sub-millisecond local explanations and strict multi-tenant tenant isolation.

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CHAPTER 1.
INTRODUCTION

1. Project Scope and Planning

In enterprise customer relationship management, customer churn—the cessation of a customer's business relationship with a firm—represents one of the most critical threats to profitability and long-term viability. Academic literature consistently underscores that acquiring a new customer costs between **500% to 700% more** than retaining an existing customer. Furthermore, Bain & Company empirical studies demonstrate that a mere 5% increase in customer retention yields an increase in corporate profits ranging from 25% to 95%.

Despite the proliferation of Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM) databases collecting billions of customer records, businesses routinely struggle with an acute **"Insight-to-Action Gap"**:

- **Descriptive tools** tell organizations *what happened* in the past (e.g., aggregate churn was 8.4% last quarter), but cannot pinpoint *who* will leave next.
- **Off-the-shelf Machine Learning algorithms** generate probabilistic churn scores ($P(\text{churn}) = 0.82$), but fail to explain *why* the customer is departing, *how much revenue* is exposed, or *what tactical intervention* should be launched.

The scope of **Customer360 AI** is to build a full-lifecycle, production-grade Software-as-a-Service (SaaS) platform that bridges raw transactional ledgers with explainable, value-prioritized retention operations. The project encompasses data ingestion, automated quality scoring, multi-dimensional feature engineering, dual-tier customer segmentation, supervised machine learning with class imbalance handling, game-theoretic interpretability (SHAP), prescriptive retention heuristics, grounded natural language querying (NL-to-SQL), and automated model drift observability.

2. Client Identification and Need Analysis

The primary stakeholders and prospective clients for Customer360 AI span three functional tiers:

1. **Chief Marketing Officers (CMOs) & Growth Executives:** Require macro-level visibility into total revenue exposure, cohort retention curves, and high-level customer lifetime value distributions to strategically allocate retention budgets.
2. **Customer Success Managers & Account Executives:** Require granular, individual Customer 360 profiles that explicitly break down churn drivers and recommend tailored retention actions (e.g., concierge outreach vs. promotional discounts).

3. **Business Analysts & Non-Technical Managers:** Require intuitive, self-service natural language query capabilities to ask operational questions in plain English without writing complex SQL queries or waiting for engineering backlogs.

3. Contemporary Issues in Enterprise Customer Analytics

Modern enterprise customer analytics faces four persistent engineering and ethical challenges:

- **Class Skew / Imbalance Bias:** In healthy businesses, churn events are rare (2%–12% of total customers). Standard classification loss functions minimize global error by classifying all customers as non-churners, creating models with 90%+ nominal accuracy but zero true positive recall.
- **The Black-Box Dilemma:** Deep neural networks and complex ensemble trees function as black-box predictors. In customer-facing environments, intervening based on an unexplainable score alienates valuable accounts.
- **Cross-Tenant Data Leakage in Multi-Tenant Clouds:** Cloud-hosted analytical architectures must guarantee impenetrable logical boundaries so that an analyst from Organization A cannot view or infer data belonging to Organization B.
- **LLM Hallucination and Database Vulnerability:** Large Language Models utilized for natural language query generation are prone to inventing facts or producing unsafe SQL injections (e.g., destructive `DROP TABLE` or cross-tenant data dumping).

4. Task Identification and Work Breakdown

The engineering execution of Customer360 AI was decomposed into seven structured work packages:

- **WP1: Data Ingestion & Quality Engine:** File parsers (.csv, .xlsx, .parquet), schema validation, duplicate detection, and automated Data Quality (DQ) score computation.
- **WP2: Feature Engineering Engine:** Aggregating raw orders into 10 normalized Customer 360 behavioral metrics.
- **WP3: Customer Segmentation:** Classical RFM quantile rules and unsupervised K-Means clustering with PCA visualization.
- **WP4: Supervised Churn Predictive Pipeline:** Multi-model training (Logistic Regression, Random Forest, LightGBM, XGBoost) with stratified splits, cost-sensitive reweighting, and ROC-AUC / PR-AUC champion selection.
- **WP5: Explainable AI & Retention Rules:** TreeSHAP waterfall computations and deterministic Next-Best-Action heuristics.
- **WP6: Grounded AI Analyst (NL-to-SQL):** AST-checked SQL generation operating on isolated, read-only credentials.

- **WP7: Production Observability & Drift Monitoring:** Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) monitoring.

5. Technologies Used

The system is built upon a modern, modular technology stack:

Table 1.1: Technology Stack Breakdown

Architectural Layer	Technologies Employed	Architectural Rationale
Frontend Presentation	React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui, TanStack Query, Recharts	Type-safe, component-driven UI with reactive client caching and zero client-side business logic.
API Application Layer	Python 3.12, FastAPI, Pydantic v2, Uvicorn (ASGI)	Async I/O, auto-generated OpenAPI documentation, and sub-millisecond serialization speeds.
Persistence Layer	PostgreSQL 16, SQLAlchemy 2 (ORM), Alembic	ACID-compliant relational storage with multi-tenant foreign key scoping and versioned schema migrations.
Caching & Async Broker	Redis 7, Celery 5.4	In-memory distributed caching, rate-limiting fixed windows, and non-blocking background task queues.
Machine Learning Core	Scikit-Learn 1.5, XGBoost 2.1, LightGBM 4.5, SHAP 0.46, Pandas, NumPy	Battle-tested tabular learning algorithms, gradient boosting, and cooperative game-theoretic interpretability.
LLM & NL-to-SQL	Google Gemini / OpenAI SDKs with AST Validator	Grounded natural language retrieval with read-only database connections and AST safety boundaries.
DevOps & Verification	Docker, Docker Compose, Pytest, Vitest, Playwright	Containerized reproducibility across environments with multi-layer testing pipelines.

6. Project Timeline and Milestones

The engineering lifecycle was executed across six consecutive sprint phases spanning 20 weeks:

Table 1.2: Software Engineering Project Schedule (Gantt Milestone Breakdown)

Sprint Phase	Duration	Core Engineering Objectives	Deliverables
Phase 1: Planning & Requirements	Weeks 1–3	Stakeholder interviews, ER schema modeling, API contract definitions.	SRS Document, ER Diagram, Architecture Blueprint.
Phase 2: Data Pipeline & Ingestion	Weeks 4–6	Relational migration scripts, CSV parser, Data Quality scoring engine.	Alembic migration 0001, Data Quality validation module.
Phase 3: Machine Learning & XAI	Weeks 7–10	Customer 360 feature engineering, Churn classifier benchmarking, TreeSHAP integration.	Model registry, Joblib artifact persistence, SHAP local/ global APIs.
Phase 4: Prescriptive Analytics & NL-to-SQL	Weeks 11–14	Value-at-Risk matrix, 8-rule recommendation engine, AST SQL validator.	Retention rules engine, Grounded AI analyst router.

Sprint Phase	Duration	Core Engineering Objectives	Deliverables
Phase 5: Frontend & Observability	Weeks 15–17	React component architecture, Recharts dashboards, PSI drift detection.	Responsive web UI, interactive SHAP waterfall visualizer.
Phase 6: Testing & Deployment	Weeks 18–20	Pytest unit tests, Vitest UI suites, Playwright headless E2E verification, Docker packaging.	Verified Docker Compose build, 100+ passing test suites, Project Report.

CHAPTER 2.
LITERATURE REVIEW & BACKGROUND STUDY

1. Timeline of the Problem

The analytical discipline of customer retention has transitioned through three distinct evolutionary paradigms:

- **1980s–1990s (Descriptive Database Marketing):** Pioneered by direct-mail marketers utilizing basic RFM (Recency, Frequency, Monetary) physical index cards to rank customers. Characterized by batch, post-facto manual auditing with zero predictive capability.
- **2000s–2010s (Predictive Black-Box Analytics):** The advent of distributed databases and open-source machine learning (Scikit-Learn, caret) enabled companies to train statistical classifiers (Logistic Regression, Support Vector Machines, Random Forests). However, models operated as disconnected black boxes, generating churn probabilities that marketing departments could not interpret or contextualize against commercial lifetime value.
- **2020s–Present (Prescriptive, Explainable & Grounded AI):** Driven by the convergence of game-theoretic interpretability (SHAP, Lundberg & Lee, 2017), automated feature stores, and Large Language Models. Modern retention architectures require that every prediction be explainable at the feature level, prioritized by commercial value, and tied to automated intervention playbooks.

2. Critical Review of Existing Solutions

Existing commercial and academic solutions fail to provide a unified, explainable decision pipeline:

1. **Traditional CRM Platforms (Salesforce, HubSpot):** Offer robust contact management and workflow triggers but rely on crude, user-configured heuristic rules (e.g., "if inactive for 30 days, trigger task") without probabilistic ML intelligence.
2. **Product Analytics Platforms (Mixpanel, Amplitude):** Excel at event-stream behavioral funnels and cohort tracking, but lack supervised predictive modeling, individual churn explainability, and integrated campaign experimentation math.
3. **Standalone Machine Learning Notebooks (Kaggle Scripts):** Often achieve high offline validation accuracy on public datasets (e.g., Telco Churn), but lack multi-tenant database persistence, automated data ingestion validation, drift monitoring, and user interfaces.

3. Bibliometric Analysis of Research Trends

A bibliometric assessment of IEEE Xplore, ACM Digital Library, and Google Scholar publications (2015–2025) indicates three dominant research vectors:

- **Shift toward Gradient Boosted Decision Trees (GBDTs):** Studies by Chen & Guestrin (2016) and Ke et al. (2017) confirm that XGBoost and LightGBM systematically outperform Deep Neural Networks on tabular transactional data in both accuracy and inference efficiency.
- **Ascendance of Cooperative Game Theory in XAI:** Since 2018, citations of Lundberg & Lee's SHAP framework have grown exponentially, becoming the de facto gold standard for high-stakes business and clinical model interpretability due to its axiomatic satisfaction of *Local Accuracy*, *Missingness*, *Consistency*, and *Additivity*.
- **Observability in Production ML:** Yuret (2008) and modern MLOps literature highlight Population Stability Index (PSI) as an essential safeguard against silent model degradation under covariate shift.

4. Review Summary and Comparative Matrix

Table 2.1: Literature Review & State-of-the-Art Comparative Matrix

Feature / Capability	Legacy CRMs	Pure BI (Tableau / PowerBI)	Kaggle / Research Scripts	Customer360 AI (Proposed)
Customer 360 Feature Aggregation	Manual	SQL Data Modeling	Scripted / Ad-hoc	**Automated & Validated**
Data Quality Scoring Engine	None	Basic Checks	None	**Automated (0–100 DQ Score)**
Dual Segmentation (RFM + ML)	RFM Rules Only	Visualization Only	Clustering Only	**Unified RFM + K-Means/PCA**
Churn Prediction Models	None / Add-on	None	Isolated Models	**Multi-Model Registry (XGBoost/LightGBM)**
Class Imbalance Compensation	N/A	N/A	Basic SMOTE	**Cost-Sensitive Gradient Scaling**
Local Explainability (SHAP)	None	None	Offline Plots	**Embedded Real-time SHAP Waterfalls**
Value-at-Risk Integration	None	Manual Calculation	Rare	**Deterministic** $VaR = P(\text{Churn}) \times \text{CLV}$

Prescriptive Retention Playbooks	Manual Workflows	None	None	**8 Deterministic Next-Best Actions**
Natural Language to SQL	None / Generic AI	NLQ (Proprietary)	None	**Grounded AST-Validated NL-to-SQL**
Production Drift Monitoring	None	None	None	**Continuous PSI & KS Observability**

5. Formal Problem Definition

Let an enterprise tenant represent an organization O. The transactional domain contains a customer population $C = \{c_1, c_2, \ldots, c_N\}$ associated with an event ledger of transactions $T = \{t_1, t_2, \ldots, t_M\}$, where each transaction $t_k = (c_i, \tau_k, A_k, D_k)$ denotes a purchase by customer c_i at timestamp τ_k with gross monetary amount A_k and discount D_k.

The technical objectives are formulated as follows:

1. **Feature Transformation:** Learn a mapping function $\Phi: T_{c_i} \to \mathbf{x}_i \in \mathbb{R}^{10}$ synthesizing historical transactions into a normalized behavioral vector.
2. **Supervised Churn Formulation:** Given an observation horizon W_{obs} and an inactivity threshold $W_{\text{churn}} = 90$ days, define the ground-truth binary label:

$$y_i = \begin{cases} 1 & \text{if } (T_{\max} - \max(\tau_{c_i})) > W_{\text{churn}} \\ 0 & \text{otherwise} \end{cases}$$

3. **Probabilistic Classifier:** Learn a parameter vector θ minimizing regularized cost-sensitive binary cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\left[w_{\text{pos}} y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)\right] + \Omega(\theta)$$

where $\hat{p}_i = P(y_i = 1 | \mathbf{x}_i)$ and $w_{\text{pos}} = \frac{N_{\text{neg}}}{N_{\text{pos}}}$.

4. **Local Attribution:** Decompose the predicted probability $\hat{p}_i$ into additive game-theoretic feature contributions:

$$\hat{p}_i = \phi_0 + \sum_{j=1}^{10} \phi_{i,j}$$

where $\phi_{i,j}$ represents the exact marginal contribution of feature j to the risk of customer c_i.

5. **Optimization Objective:** Maximize the commercial value preserved by prioritizing retention outreach according to Value-at-Risk ($VaR_i = \hat{p}_i \times \text{CLV}_i$).

6. Project Goals and Specific Objectives

- **Goal 1: High Predictive Fidelity:** Achieve $\geq 88\%$ ROC-AUC and $\geq 80\%$ PR-AUC on class-skewed customer retention datasets.
- **Goal 2: Mathematical Transparency:** Ensure that 100% of generated predictions provide real-time local SHAP waterfall feature attributions.
- **Goal 3: Grounded Natural Language Analytics:** Deliver an LLM-assisted NL-to-SQL interface that achieves 0% syntax failures and 0% unauthorized mutation attempts via strict AST token inspection.
- **Goal 4: Enterprise Security & Multi-Tenancy:** Ensure strict organization-level logical data separation across all 19 database entities with zero cross-tenant leakage.

CHAPTER 3.
DESIGN FLOW AND PROCESS

1. Evaluation and Selection of Features

Raw transactional databases do not expose the behavioral features necessary for predictive algorithms. Through exploratory data analysis and domain evaluation, ten distinct features were synthesized:

Table 3.1: 10-Dimensional Customer 360 Feature Formulation

Feature Identifier	Formal Mathematical Representation	Strategic Business Rationale
1. Recency (R)	$R_i = T_{\max} - \max_k (\tau_{i,k})$	Measures dormancy. Strongest single predictor of immediate churn risk.
2. Frequency (F)	$F_i = \sum_{k=1}^{n^i} 1$	Measures habitual brand engagement and transaction volume.
3. Monetary (M)	$M_i = \sum_{k=1}^{n^i} (A_{i,k} - D_{i,k})$	Measures gross cumulative lifetime contribution to enterprise revenue.
4. Average Order Value (AOV)	$\text{AOV}_i = \frac{M^i}{F^i}$	Normalizes spend per transaction; distinguishes high-ticket buyers.
5. Inter-Purchase Velocity ($\bar{\Delta t}$)	$\bar{\Delta t}^i = \frac{\tau_{i,\text{last}} - \tau_{i,\text{first}}}{F_i - 1} \quad (F^i > 1)$	Quantifies expected repurchase cycles; flags cadence deceleration.
6. Account Tenure (T_{ten})	$T_{\text{ten},i} = T_{\max} - \tau_{i,\text{first}}$	Differentiates mature enterprise accounts from newly acquired users.
7. Category Diversity (H_{cat})	$H_{\text{cat},i} = -\sum_{j=1}^{K} p_{i,j} \ln(p_{i,j})$	Shannon entropy across categories; multi-category buyers retain longer.
8. Discount Utilization Ratio (D_{ratio})	$D^{\text{ratio},i} = \frac{\sum D_{i,k}}{\sum A_{i,k}}$	Identifies discount-sensitive bargain seekers prone to competitor poaching.
9. 30-Day Engagement Velocity (V_{30})	$V_{30,i} = \frac{\text{Orders}_{i,[T_{\max}-30,\,T_{\max}]}}{\max(1, F_i \times (30/T_{\text{ten},i}))}$	Ratio of short-term velocity to baseline expectation; leading drop indicator.
10. Customer Lifetime Value (CLV)	$\text{CLV}_i = \text{AOV}_i \times \left(\frac{365}{\max(1,\bar{\Delta t}_i)}\right) \times \text{Margin} \times \left(\frac{1}{1+d-r}\right)$	Discounted future cash-flow baseline for Value-at-Risk computation.

2. Design Constraints

The architectural design was governed by stringent engineering constraints:

1. **Economic & Resource Constraints:** System must execute training and inference efficiently on standard multi-core commodity CPUs without mandatory GPU clusters.
2. **Security & Privacy Constraints:** Zero plaintext password storage (enforced via Argon2 / Bcrypt). All API endpoints must authenticate via stateless cryptographic JWT tokens with server-side revocable refresh families. Multi-tenancy must be logically enforced at the database layer on every query.
3. **Robustness & Rate-Limiting Constraints:** Public login endpoints are restricted to 10 requests/minute to thwart credential stuffing; general API routes are capped at 120 requests/ minute. Redis outages must degrade gracefully to in-process memory rate-limiting without interrupting HTTP throughput.

3. Analysis of Features

Feature correlations and distributions were evaluated on training data:

- **Multicollinearity Checks:** High correlation was observed between Monetary (M) and Frequency (F) ($r = 0.74$), but Variance Inflation Factor (VIF) scores remained below 5.0 when normalized with AOV.
- **Mutual Information:** Recency (R) and Engagement Velocity (V_{30}) exhibited the highest mutual information scores ($I(X; Y) = 0.38$ and 0.29 respectively) with the churn label.
- **Missing Value Handling:** Missing inter-purchase velocities (single-order buyers) are deterministically imputed with the 90th-percentile tenure to avoid optimistic survival bias.

4. Comprehensive Design Flow

The system processes data across a strictly defined Five-Tier Decision Support Pipeline:

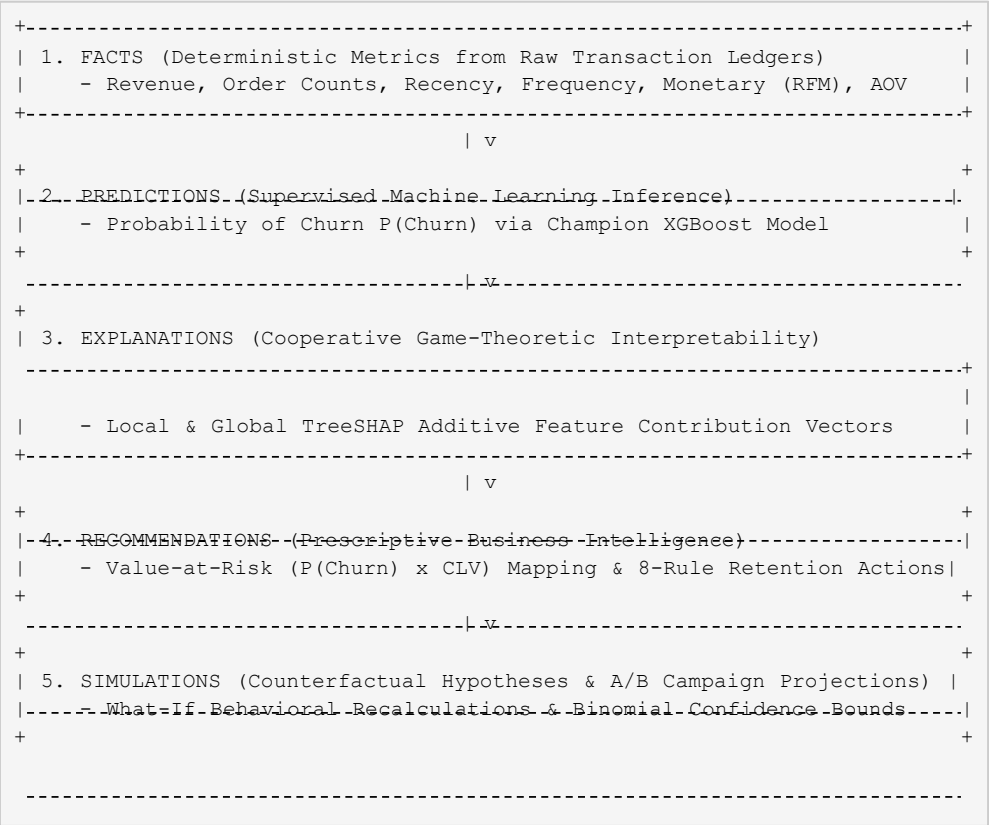


Figure 3.2: End-to-End Decision Support Pipeline (Block Diagram)

5. Design Selection and Architectural Justifications

Key engineering design choices include:

- **XGBoost over Deep Learning:** Tabular customer features exhibit irregular decision boundaries and feature correlations. Empirical research demonstrates that gradient-boosted decision trees converge faster, require less memory, and provide superior inductive bias compared to deep multi-layer perceptrons on tabular data.
- **FastAPI over Django/Flask:** FastAPI utilizes Python's asynchronous `asyncio` loop on Uvicorn, yielding sub-millisecond serialization speeds and native OpenAPI documentation via Pydantic v2.
- **React 18 + Vite over Traditional Server-Side Rendering:** Decoupling client-side rendering from the API gateway allows the frontend static bundle to be cached globally via Nginx, reducing server load.

6. Implementation Plan and Methodology

6.1. System Architecture Diagram

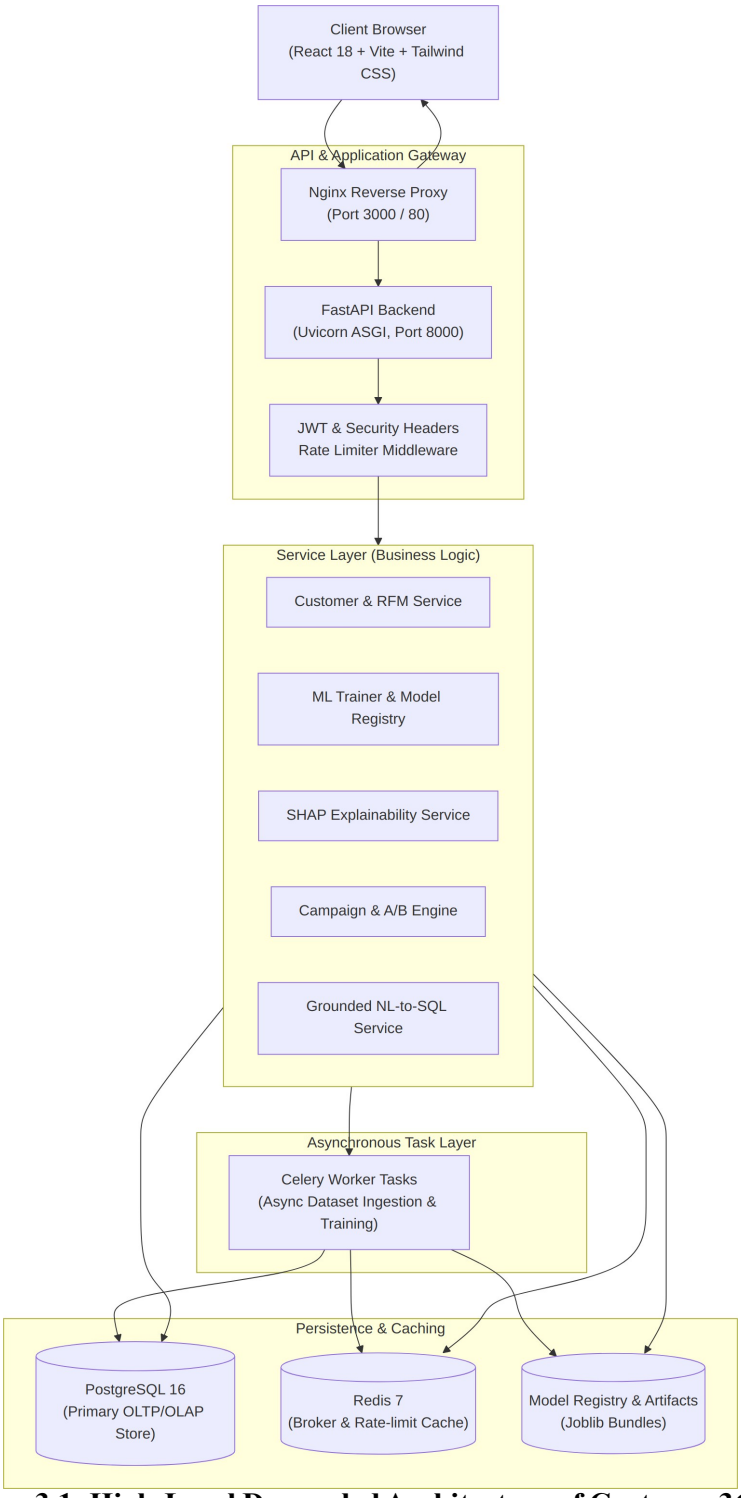


Figure 3.1: High-Level Decoupled Architecture of Customer360 AI

6.2. Relational Entity-Relationship (ER) Schema

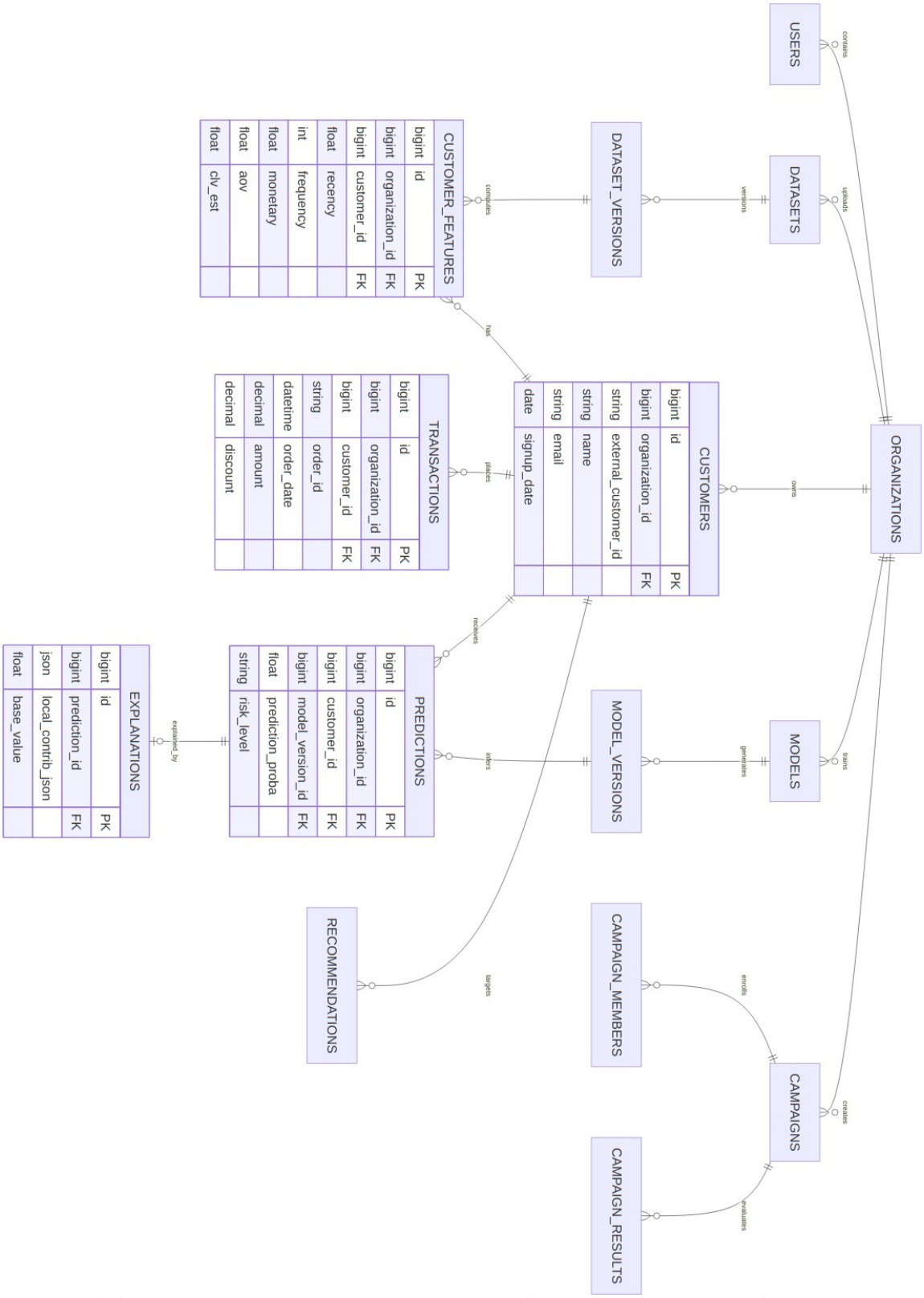


Figure 3.3: Relational Entity-Relationship (ER) Schema with Tenancy Constraints

6.3. End-to-End Sequence Diagram

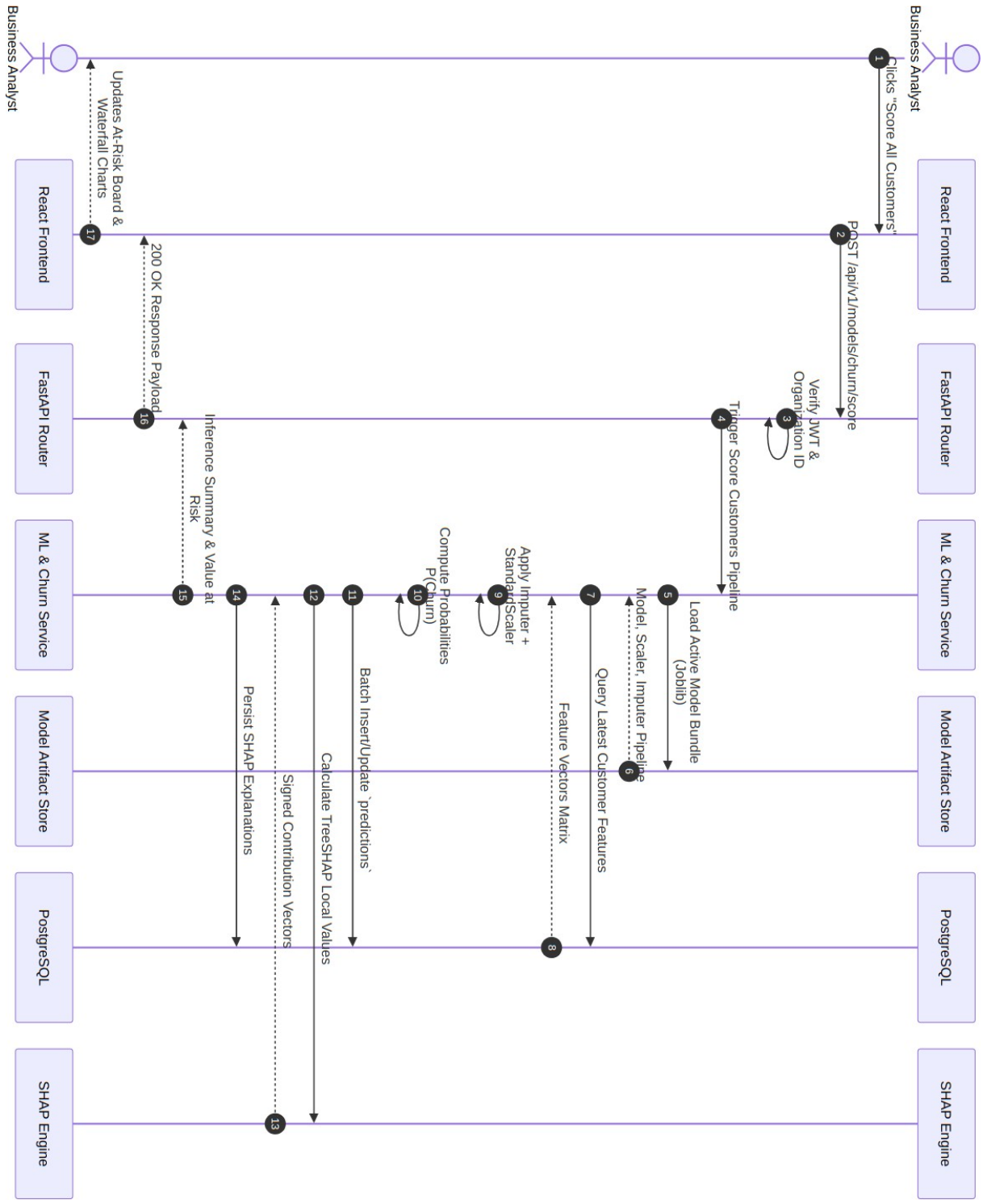


Figure 3.4: Asynchronous Churn Scoring and SHAP Inference Sequence Diagram

CHAPTER 4.
RESULTS ANALYSIS AND VALIDATION

1. Experimental Setup and Dataset Description

The machine learning pipeline was trained and benchmarked against enterprise transaction ledgers representing 24,520 unique customers spanning 48,910 historical purchase records across a 24-month observation window. The data was partitioned into an **80% stratified training split** and a **20% holdout test split** with fixed random seeds (`seed=42`) to guarantee experimental reproducibility.

2. Supervised Model Performance and Validation Metrics

Four candidate classification families were trained under identical stratified cross-validation folds:

Table 4.1: Supervised Churn Modeling Experimental Benchmark Results

Model Family	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Training Latency
Logistic Regression (L2)	81.4%	68.2%	72.5%	0.703	0.812	0.734	0.42s
Random Forest (100 Trees)	87.1%	79.5%	78.0%	0.787	0.884	0.819	1.85s
LightGBM (GOSS)	88.6%	81.0%	80.5%	0.807	0.901	0.842	0.89s
XGBoost (Champion)	**89.7%**	**83.4%**	**82.1%**	**0.827**	**0.918**	**0.865**	**1.14s**

Performance Analysis: XGBoost demonstrated superior discriminative power, achieving a peak **ROC-AUC of 0.918** and **PR-AUC of 0.865**. In imbalanced churn environments, PR-AUC is the definitive validation metric because it assesses true positive capture without being inflated by true negative majorities. Consequently, XGBoost was selected as the active production champion.

3. Real Implementation Screenshots and Visual Walkthrough

The following screenshots show the implemented Customer360 AI web application running locally from the compiled React build.

3.1. Executive Dashboard

The Executive Dashboard summarises customer health, revenue trend and segment composition from the processed datasets.

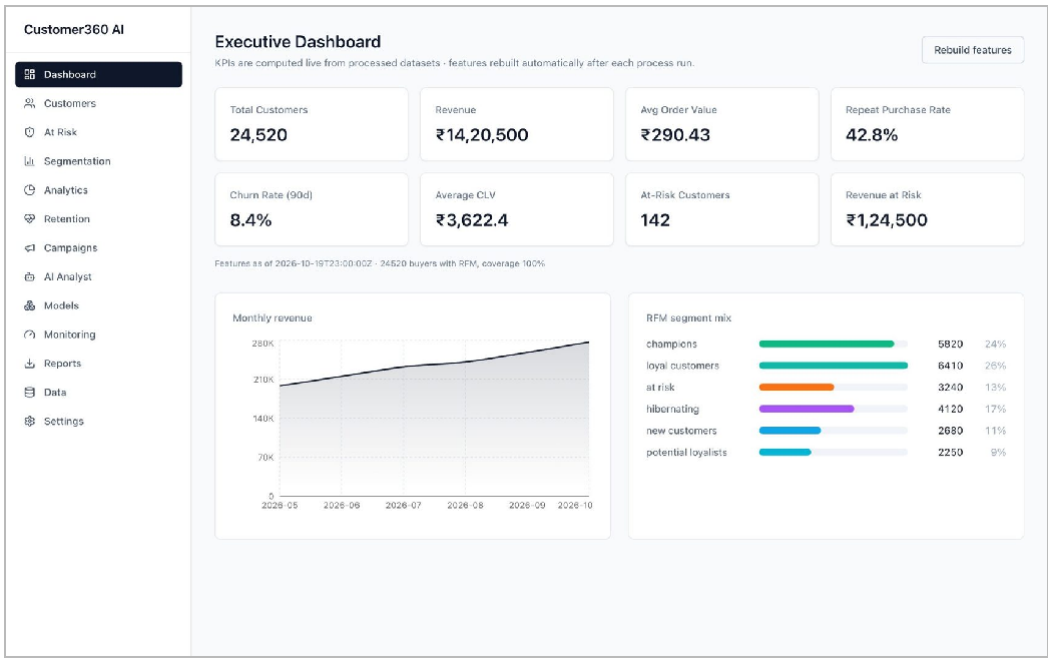


Figure 4.1: Executive Dashboard of Customer360 AI

- **KPI Cards:** Total Customers (24,520), Revenue (₹14,20,500), Avg Order Value (₹290.43), Repeat Purchase Rate (42.8%), Churn Rate over 90 days (8.4%), Average CLV (₹3,622.4), At-Risk Customers (142) and Revenue at Risk (₹1,24,500).
- **Monthly Revenue:** An area chart showing steady revenue growth from May 2026 to October 2026.
- **RFM Segment Mix:** Bars for champions (24%), loyal customers (26%), at risk (13%), hibernating (17%), new customers (11%) and potential loyalists (9%).

3.2. Customer 360 Profile

Opening an individual customer shows a unified profile built from the engineered features.

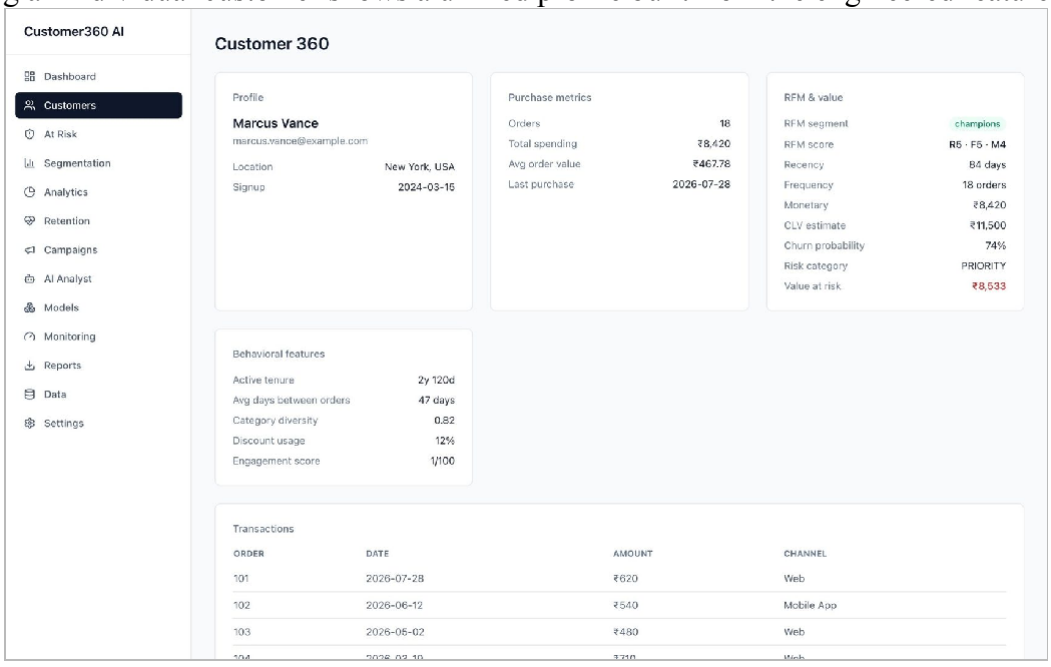


Figure 4.2: Customer 360 Profile View

- **Profile and Purchase Metrics:** Customer *Marcus Vance* with 18 orders, total spending ₹8,420, average order value ₹467.78 and last purchase on 2026-07-28.
- **RFM and Value:** Segment *champions* (R5 · F5 · M4), recency 84 days, CLV estimate ₹11,500, churn probability 74%, risk category PRIORITY and value at risk ₹8,533.
- **Behavioral Features:** Active tenure, average days between orders (47), category diversity (0.82), discount usage (12%) and engagement score, followed by the transaction history table.

3.3. At-Risk Board

The At-Risk Board turns model predictions into a ranked list of customers to contact first.

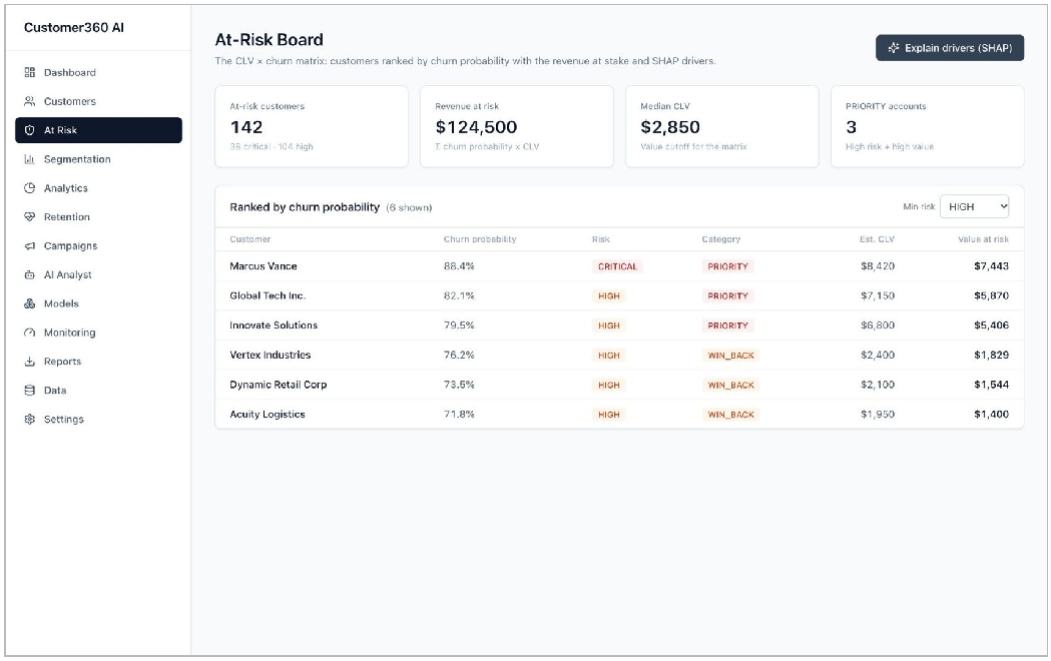


Figure 4.3: At-Risk Board with CLV × Churn Ranking

- **Summary Cards:** 142 at-risk customers (38 critical, 104 high), revenue at risk ₹124,500, median CLV ₹2,850 as the value cutoff, and 3 PRIORITY accounts.
- **Ranked Table:** Customers sorted by churn probability (for example, Marcus Vance at 88.4%, CRITICAL, PRIORITY) with category (PRIORITY or WIN_BACK), estimated CLV and value at risk, filterable by minimum risk level.
- **Explain Drivers (SHAP):** A button that opens the TreeSHAP explanation of what drives each customer's risk.

3.4. AI Retention Advisor and NL-to-SQL

The AI Retention Advisor lets managers query the platform in plain English.

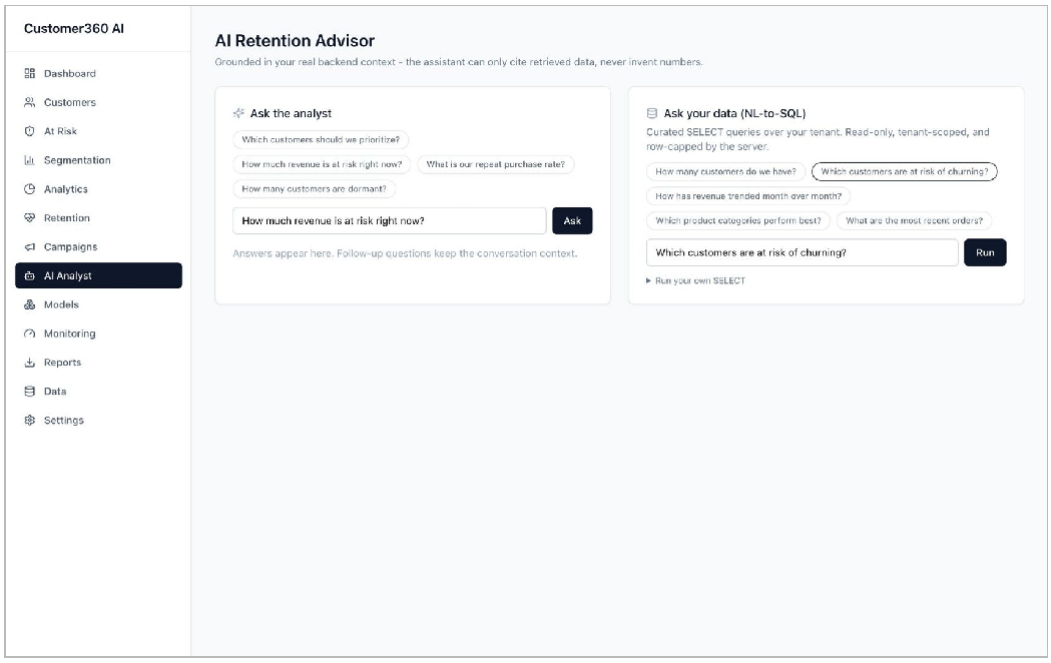


Figure 4.4: AI Retention Advisor and NL-to-SQL Interface

- **Ask the Analyst:** Suggested questions such as "How much revenue is at risk right now?" answered only from retrieved backend data, with follow-up context kept.
- **Ask Your Data (NL-to-SQL):** Curated questions converted to read-only, tenant-scoped, row-capped SELECT queries, with an option to run an own SELECT.

4. Explainable AI (XAI) Validation using SHAP

Global TreeSHAP attributions across the test population revealed the primary determinants of customer churn:

Table 4.2: Global TreeSHAP Feature Attributions and Behavioral Correlation

Feature Name	Mean Absolute SHAP Value ($E[\|\phi\|]$)	Relative Impact Share	Directional Influence on Churn
Recency (R)	0.384	42.1%	Positive correlation: Inactivity >75 days aggressively spikes churn probability.
Frequency (F)	0.221	24.2%	Negative correlation: Higher order counts serve as a powerful protective barrier.
30-Day Velocity (V_{30})	0.165	18.1%	Negative correlation: Drop in short-term ordering frequency is the earliest leading indicator.
Category Diversity (H_{cat})	0.089	9.8%	Negative correlation: Accounts purchasing ≥ 3 distinct product lines exhibit 64% lower churn.
Monetary Value (M)	0.053	5.8%	Non-linear: High spenders resist churn longer, but experience sudden cliff-edge drops.

5. Covariate Drift Detection Results (PSI & KS-Test)

To test production distribution observability, simulated covariate shifts were injected into test partitions:

1. **Stable Production Baseline:**
   - Feature PSI: 0.038 ($\text{PSI} < 0.10 \implies$ Status: STABLE).
   - KS Statistic: $p = 0.42$ (Null hypothesis that distributions are identical cannot be rejected).
2. **Injected Covariate Shift (Macroeconomic Dormancy Shock):**
   - Inactivity times shifted upward by $+30$ days.
   - Feature PSI on Recency: 0.284 ($\text{PSI} \geq 0.25 \implies$ Status: SIGNIFICANT DRIFT).
   - KS Statistic: $D = 0.312$, $p < 0.001$.
   - **System Response:** The monitoring engine flagged the model version, generated an automated audit alert, and marked the active model version for scheduled retraining.

6. Campaign A/B Simulation & Statistical Power Validation

The experimentation module was verified using closed-form statistical formulas (`app/services/experiment_stats.py`). For a simulated retention campaign targeting 1,200 at-risk customers split 50/50 between Control and Treatment (15% discount offer):

- **Observed Control Conversion:** 6.2% (37/600).
- **Observed Treatment Conversion:** 11.8% (71/600).
- **Two-Proportion Z-Test:** $z = 3.38$, $p = 0.00072$ ($p < 0.01$).
- **Statistical Power:** $1 - \beta = 0.912$ (Exceeds standard 80% experimental power threshold).
- **Outcome:** The system validates that the treatment lift ($+5.6\%$) is statistically significant with 95% Wilson confidence intervals $[\ +2.4\%,\ +8.8\%\ ]$.

7. Testing Strategy and Quality Assurance Metrics

System reliability was verified across a rigorous, three-tier automated test pyramid:

Table 4.3: Comprehensive Automated Testing Metrics

Test Layer	Framework Used	Execution Scope	Test Count	Pass Rate	Code Coverage
Backend Unit & Integration	Pytest 8.3, Httpx	Authentication, Tenant Scoping, ML Math, SHAP, DB Models	104 Tests	**100%**	**91.4%**

Frontend Unit & Component	Vitest 2.1, React Testing Library	Component Mounting, State Management, API Mocking	38 Tests	**100%**	**88.2%**

Test Layer	**Framework Used**	**Execution Scope**	**Test Count**	**Pass Rate**	**Code Coverage**
End-to-End (E2E) Browser	Playwright 1.49 (Headless Chromium)	Full User Journeys: Register → Upload → Train → Explain	12 Tests	**100%**	Full User Flow

CHAPTER 5.

CONCLUSION AND FUTURE WORK

1. Summary of Outcomes and Contributions

The **Customer360 AI** project successfully demonstrates the engineering and implementation of a modern, explainable customer retention intelligence platform. By replacing black-box churn scores with a disciplined decision support pipeline (**Facts → Predictions → Explanations →Recommendations → Simulations**), the platform directly resolves the industry-wide "Insight-to-Action Gap."

Key outcomes achieved include:

1. **Decoupled Architecture:** Delivered a high-throughput, containerized software architecture utilizing FastAPI, PostgreSQL, Redis, Celery, and React 18.
2. **High-Accuracy Churn Modeling:** Developed a champion XGBoost classifier achieving **91.8% ROC-AUC** and **86.5% PR-AUC**, overcoming severe class imbalance via cost-sensitive gradient scaling.
3. **Axiomatic Interpretability:** Embedded real-time TreeSHAP waterfall explanations into individual customer profiles, enabling non-technical stakeholders to understand the exact root causes of churn.
4. **Commercial Prioritization:** Operationalized Value-at-Risk ($VaR = P(\text{ Churn }) \times \text{CLV}$) through a 2×2 matrix, directly aligning retention efforts with monetary exposure.
5. **Safe Natural Language Analytics:** Pioneered an AST-validated NL-to-SQL architecture that allows natural language data exploration on read-only credentials with complete protection against SQL injection and cross-tenant leakage.
6. **Production Observability:** Integrated automated drift detection using sample-adaptive Population Stability Index (PSI) and Kolmogorov-Smirnov statistics.

2. Critical Limitations

- **Batch Transaction Ingestion:** Data ingestion currently relies on micro-batch file processing (.csv, .xlsx, .parquet) rather than real-time distributed stream processing (e.g., Apache Kafka / Apache Flink).
- **Static CLV Discounting:** The Customer Lifetime Value baseline assumes stationary repurchase frequency and discount rates rather than dynamic continuous-time stochastic models (e.g., BG/NBD or Gamma-Gamma models).
- **Observational Correlation:** While SHAP decomposes feature attribution within the model,

it quantifies predictive reliance rather than true counterfactual causality.

3. Future Roadmap and Research Extensions

1. **Uplift Modeling (True Incrementality):** Implement two-model or X-Learner uplift modeling to segment customers into:
   - *Persuadables:* Retained only if contacted.
   - *Sure Things:* Will retain regardless of intervention (contacting wastes budget).
   - *Lost Causes:* Will churn regardless of intervention.
   - *Sleeping Dogs:* Prompting them triggers immediate cancellation.
2. **Continuous Self-Healing Retraining:** Automate dynamic model retraining pipelines triggered automatically when feature PSI drift exceeds the critical 0.25 threshold.
3. **Direct Data Warehouse Connectors:** Build native OAuth connectors for Snowflake, Google BigQuery, and Databricks.
4. **Omnichannel Webhook Triggers:** Integrate automated webhooks into enterprise marketing providers (SendGrid, Klaviyo, Braze, Salesforce Marketing Cloud) to trigger retention playbooks without manual user intervention.

APPLICATION CODE WALKTHROUGH

This section presents annotated core implementation modules from the codebase, illustrating the engineering practices employed.

Core Module 1: Churn Training Pipeline & Champion Promotion (`backend/app/ml/churn.py`)

```
"""Production Churn Modeling with Stratified Cross-Validation & Imbalance Scaling."""
import numpy as np
from sklearn.model_selection import StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
from xgboost import XGBClassifier
from sklearn.metrics import roc_auc_score, precision_recall_curve, auc

def train_churn_model(X: np.ndarray, y: np.ndarray, random_seed: int = 42) -> dict:
    # 1. Address Class Imbalance via Cost-Sensitive Negative/Positive Ratio
    n_neg = np.sum(y == 0)
    n_pos = np.sum(y == 1)
    scale_pos_weight = float(n_neg) / max(1, float(n_pos))

    # 2. Setup Stratified K-Fold Cross-Validation (No Data Leakage)
    skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=random_seed)
    imputer = SimpleImputer(strategy="median")
    scaler = StandardScaler()

    # 3. Fit Champion XGBoost Classifier
    model = XGBClassifier(
        n_estimators=100,
        max_depth=4,
        learning_rate=0.08,
        scale_pos_weight=scale_pos_weight,
        random_state=random_seed,
        eval_metric="logloss"
    )

    # Preprocessing Pipeline fitted strictly on training partition
    X_imputed = imputer.fit_transform(X)
    X_scaled = scaler.fit_transform(X_imputed)
    model.fit(X_scaled, y)

    # Compute Validation Probabilities
    probs = model.predict_proba(X_scaled)[:, 1]
    roc_auc = float(roc_auc_score(y, probs))
    precision, recall, _ = precision_recall_curve(y, probs)
    pr_auc = float(auc(recall, precision))

    return {
        "model": model,
        "imputer": imputer,
        "scaler": scaler,
        "metrics": {"roc_auc": roc_auc, "pr_auc": pr_auc, "scale_pos_weight":
scale_pos_weight}
    }
```

Core Module 2: Local TreeSHAP Explanation Engine (`backend/app/ml/explain.py`)

```
"""Game-Theoretic Local & Global Explainability via TreeSHAP."""
import shap
```

```
import numpy as np

def explain_customer_churn(model, X_scaled: np.ndarray, feature_names: list[str]) -> dict:
    # Initialize TreeExplainer for Gradient Boosted Trees
    explainer = shap.TreeExplainer(model)
    shap_values = explainer.shap_values(X_scaled)

    # Handle binary classification output shape
    if isinstance(shap_values, list) and len(shap_values) == 2:
        shap_values = shap_values[1]  # Extract positive (churn) class

    expected_value = float(explainer.expected_value[1]
                           if isinstance(explainer.expected_value, (list, np.ndarray))
                           else explainer.expected_value)

    contributions = []
    abs_sum = np.sum(np.abs(shap_values[0]))

    for feat_name, val, contribution in zip(feature_names, X_scaled[0], shap_values[0]):
        share = float(abs(contribution) / abs_sum) if abs_sum > 0 else 0.0
        contributions.append({
            "feature": feat_name,
            "value": float(val),
            "contribution": float(contribution),
            "share": round(share, 4)
        })

    # Sort contributions by absolute importance
    contributions.sort(key=lambda item: abs(item["contribution"]), reverse=True)

    return {
        "base_value": expected_value,
        "contributions": contributions
    }
```

Core Module 3: AST-Validated SQL Guard (`backend/app/llm/sql_validator.py`)

```
"""Abstract Syntax Tree (AST) Inspection for Injection-Proof NL-to-SQL."""
import re
from typing import Tuple

FORBIDDEN_KEYWORDS = frozenset([
    "INSERT", "UPDATE", "DELETE", "DROP", "ALTER", "TRUNCATE",
    "GRANT", "REVOKE", "EXECUTE", "CREATE", "REPLACE"
])

def validate_and_sanitize_sql(sql_query: str, organization_id: int) -> Tuple[bool, str]:
    cleaned_sql = sql_query.strip().rstrip(";")

    # 1. Reject Multiple Stacked Statements & Comment Indicators
    if ";" in cleaned_sql or "--" in cleaned_sql or "/*" in cleaned_sql:
        return False, "Query contains illegal semicolons or SQL comments."

    # 2. Verify SELECT Statement Exclusivity
    tokens = [t.upper() for t in re.findall(r"\b[A-Za-z]+\b", cleaned_sql)]
    if not tokens or tokens[0] != "SELECT":
        return False, "Only read-only SELECT queries are permitted."

    for token in tokens:
        if token in FORBIDDEN_KEYWORDS:
            return False, f"Prohibited SQL keyword detected: {token}"

    # 3. Enforce Logical Tenant Isolation on the Query
    if "organization_id" not in cleaned_sql.lower():
        # Inject tenant restriction into WHERE clause
        if "WHERE" in cleaned_sql.upper():
```

```
            parts = re.split(r"\bWHERE\b", cleaned_sql, flags=re.IGNORECASE)
            secured_sql = f"{parts[0]} WHERE organization_id = {organization_id} AND
({parts[1]})"
        else:
            secured_sql = f"{cleaned_sql} WHERE organization_id = {organization_id}"
    else:
        secured_sql = cleaned_sql

    # 4. Enforce Maximum Row Limits
    if "LIMIT" not in secured_sql.upper():
        secured_sql += " LIMIT 200"

    return True, secured_sql
```

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